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The Application of Physics-Informed Neural Networks (PINNS) for Geothermal Well Thermal Recovery Post-Injection

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Abstract. This study explores the application of Physics-Informed Neural Networks (PINN) to simulate thermal recovery in geothermal wells following injectivity tests. The primary objective is to implement PINN to model well thermal recovery. Real well data from the literature, encompassing parameters such as circulation time, injectate temperature, heating-up surveys, and wellbore radius measurements, are gathered for analysis. Subsequently, governing equations and initial and boundary conditions for PINN modeling are established. The Ascencio and Horner analytical methods are employed to formulate boundary conditions for PINN. The application involves predicting natural state temperatures and temperature profiles across varying radii from the wellbore to the maximum radius of interest. However, it is crucial to acknowledge certain limitations. The iterative nature of PINN requires considerable computational resources, similar to numerical models. Furthermore, assuming vertical parameter independence may oversimplify reservoir behavior. In conclusion, despite limitations, PINN demonstrates the potential to aid thermal recovery simulations. Refining natural state temperature extrapolation is deemed essential, considering a tendency to overestimate values. Validating the model's accuracy involves refining physical parameters through backpropagation. Additionally, it is imperative to establish the model's credibility through a rigorous comparative analysis of outcomes with those obtained from established commercial software.

1. Introduction

Using geothermal resources for electricity generation necessitates drilling deep boreholes in strategically chosen thermal zones within geothermal fields [1][2]. A crucial aspect of designing such wells in geothermal projects involves determining the depth of the production casing shoe (PCS). This determination relies on obtaining undisturbed bottom-hole temperature (BHT) data. During the early stages of drilling exploration wells in geothermal development, acquiring sufficient data to determine the undisturbed BHT can be challenging [3]. The static formation temperature test (SFTT) is commonly employed to estimate the undisturbed temperature at specific depths during drilling to address this issue. This information aids in establishing the appropriate depth for setting the production casing shoe (PCS) [4].

In the geothermal development phase, a completion test is conducted immediately after the drilling process is completed. This test encompasses various surveys, including a dummy run, pressure-temperature-spinner (PTS) injection, multiple-rate injectivity test (II test), and a subsequent fall-off test (FOT), which are executed sequentially, requiring approximately 27 hours of rig time [4]. The fall-off test occurs immediately after the II test and involves recording downhole pressure at specific depths following the cessation of injection. Analyzing well pressure recovery data provides an initial estimation of rock transmissivity (kh), which correlates with the well's permeability, near-wellbore conditions, and boundary conditions [4]. Additionally, a method called temperature transient analysis (TTA), introduced by [5], utilizes transient temperature data generated during the fall-off test to estimate specific parameters from the geothermal reservoir. This complements the analysis conducted using pressure transient analysis (PTA).

[6] have developed a novel predictive model for calculating static formation temperature in geothermal wells, employing an artificial neural network (ANN) approach. The static formation temperature estimates derived from the ANN validation process demonstrated a high level of agreement ($R^2 > 0.95$) with actual data obtained from synthetic experiments and field observations. These results indicate that the new ANN model holds promise as a practical tool for reliably predicting static formation temperatures in geothermal wells. However, given that neural networks rely heavily on data, some reservations exist regarding their exclusive use when data are limited. In response, an NN that incorporates the principles of physics has been sought in various research domains, representing a significant stride toward the next generation of neural networks. A physics-informed neural network (PINN), as introduced by [7], has been developed to satisfy the conservation of mass and energy principles for predicting the spatial distribution of temperature, pressure, and permeability within natural-state hydrothermal systems. The PINN integrates the physics laws used in numerical simulations to constrain the predictions made by the neural network.

In the present study, we have employed a PINN approach to simulate thermal recovery following injection activities while considering the thermal diffusion equation in the loss function. The PINN approach offers potential benefits by facilitating inverse modeling of the thermal diffusivity coefficient and extrapolating temperature to the static formation temperature. In contrast to existing neural network approaches for static formation temperature prediction, our primary objective was to develop a PINN methodology that ensures the predicted temperatures and reservoir parameters adhere to the governing equations.

2. Construction of Physics-Informed Neural Networks

2.1 Governing Equation: Thermal Diffusion Equation

The thermal diffusion equation governs the temperature transient analysis. [8] developed the thermal diffusion equation (1), the primary expression for analyzing transient temperature, which only represents thermal conduction (no convection). Another significant work was produced by [9], who investigated the injection of fluids into boreholes and the heat transfer processes. This research was the first to attain an expression of temperature in terms of time and depth [10]. These two authors are the cornerstones of interpreting transient temperature data, similar to [11], who analyzed transient pressure data.

$$\frac{1}{D_t}\frac{\partial T}{\partial t} - \frac{\partial^2 T}{\partial r^2} - \frac{1}{r}\frac{\partial T}{\partial r} = 0 \tag{1}$$

where T is temperature in ($°C$), D_t is the reservoir rock thermal diffusivity (m^2/s) equal to $\kappa/(\rho C_p)$, κ is the overall (fluid and rock) thermal conductivity of porous media (W/m·K), and ρC_p is the overall (fluid and rock) thermal capacity of porous media or the reservoir (J/K·m^3).

We can solve equation (1) by applying the temperature boundary conditions. [12] derive the conditions using the Horner method described by Equation (2), with boundary conditions as the injection temperature (T_c) is constant at circulation period (t_c) observed in wellbore radius (r_w).

$$T(\Delta t) = T_i - m\log\left(\frac{t_c + \Delta t}{\Delta t}\right) \tag{2}$$

where $T(\Delta t)$ is the temperature at shut-in time $T(\Delta t)$ ($°C$), T_i the static formation temperature, m is the gradient per log cycle ($°C$), and Δt is the duration after shut-in that temperature was recorded.

2.2 Physics-informed Loss Function

The loss function of the PINN is described as follows:

$$LOSS = LOSS_m + LOSS_p + LOSS_{bc} \tag{3}$$

where $LOSS_m$ is the measure of the difference between the predictions and observations, $LOSS_p$ is the constraint on partial differential equations (PDE), and $LOSS_{bc}$ is the boundary condition. The physics constraint was adopted from Equation (1) for this research. Thus, we define and minimize the following form of the loss function.

$$LOSS_m = \frac{1}{N_m}\sum_{i=1}^{N_m}\left(T_{obs} - T_{pred}\right)^2 \tag{4}$$

$$LOSS_p = \frac{1}{N_p}\sum_{i=1}^{N_p}\left(\frac{1}{D_t}\frac{\partial T}{\partial t} - \frac{\partial^2 T}{\partial r^2} - \frac{1}{r}\frac{\partial T}{\partial r}\right)^2 \tag{5}$$

$$LOSS_{bc} = \frac{1}{N_{bc}}\sum_{i=1}^{N_{bc}}\left(T_{bc,true} - T_{bc,pred}\right)^2 \tag{6}$$

where T_{obs} and T_{pred} indicate the observations and predictions temperature values, respectively. N_m, N_p, and N_{bc} are the numbers of observed data, physical constraint domain samples, and boundary condition samples, respectively.

2.3 Boundary Condition

Like modeling in general, using PINN requires boundary conditions. The boundary condition used is the farthest radius boundary condition. It will be assumed to have a static temperature at the farthest radius. The cooling distribution radius due to temperature during injection has been determined before modeling the thermal recovery of the well. The temperature value at the specified radius is a static formation temperature obtained from analytical methods. Figure 1 shows the SFT estimation scheme for boundary conditions.

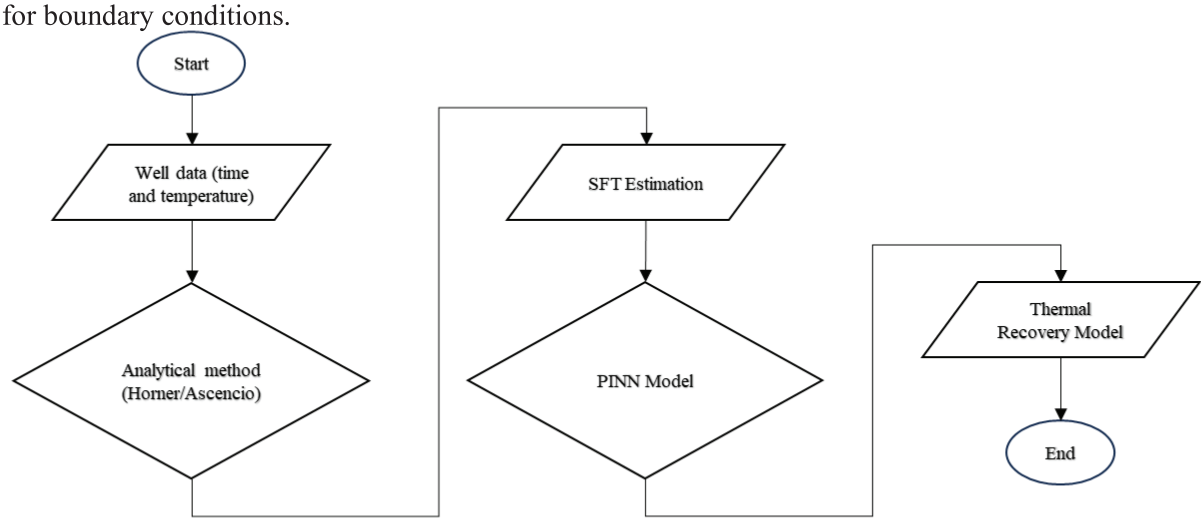

Figure 1. SFT estimation scheme.

2.4 PINN Architecture

The selection of the PyTorch framework is deliberate, as it allows for the thermal diffusivity coefficient to be treated as a learnable parameter. This flexibility enhances our model's capacity to adapt and refine the thermal diffusivity during training within the defined architecture. Inputs are specified as coordinates, encompassing the radial parameter (r) and temporal variable (t), while the chosen network configuration comprises 20 hidden neurons distributed across 8 hidden layers. The optimization strategy employed is the Adaptive Moment Optimization (Adam) algorithm, featuring a learning rate established at 1.0×10^{-3}. Notably, model training is conducted over 15,000 epochs to ensure robust convergence and performance enhancement. The PINN architecture can be seen in Figure 2.

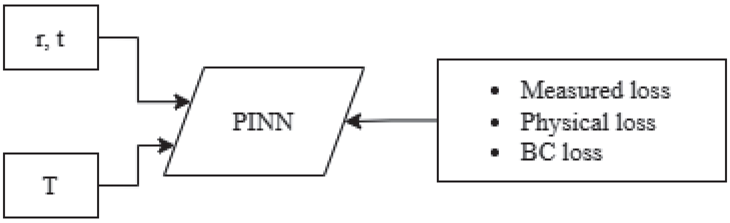

Figure 2. The architecture of PINN.

3. Validating of the Physics-Informed Neural Networks

The temperature dataset employed in this study originates from research conducted by [6] in their investigation of static formation temperature estimation in geothermal wells using an artificial neural network methodology. The thermal recovery data utilized in [6] is derived from diverse geothermal fields, including: (1) Los Humeros, Mexico [13]; (2) Oklahoma, USA [14]; (3) Mississippi, USA [15]; (4) Leyte, Philippines [16]; (5) Kyushu, Japan [17]; (6) Chipilapa, El Salvador [18][19]; (7) Lardarello, Italy [20]; and (8) Norton Sound field, Alaska [21].

4. Result and Discussion

4.1 Initial Estimation of Static Formation Temperature: Comparing the Horner Method and the Ascencio Method

Analytical solutions to estimate the value of static formation temperature are carried out using Horner's and Ascencio's methods. The use of Horner's method can be supported by the available circulation period data. The equation for the Horner method can be seen in Equation (2). The Ascencio method uses the equation below.

$$T(t) = T_i + \frac{1}{\sqrt{t}} m \tag{7}$$

where $T(t)$ is the temperature at the shut-in time ($°C$), T_i the static formation temperature, m is the gradient ($s^{\frac{1}{2}}°C$), and t is the observed period that temperature was recorded.

The results of the static formation temperature estimation can be seen in Table 1. Based on the table, the estimation results using the Ascencio method succeeded in estimating the SFT more optimistically than the Horner method. The R^2 value, which indicates the goodness of fit or how well the predicted values match the actual data, shows that the Ascencio method slightly outperforms the Horner method in terms of accuracy. A higher R^2 value suggests a stronger correlation between the observed data and the predicted results. The R^2 value is calculated using the following equation.

$$R^2 = 1 - \frac{\Sigma(\mathrm{y_i} - \hat{y_i})^2}{\Sigma(\mathrm{y_i} - \bar{y})^2} \tag{8}$$

where y_i represent the actual observed data, $\hat{y_i}$ the predicted values, and $\bar{y}$ the mean of the observed data.

The SFT estimation results from the two methods will be used as boundary conditions in modeling thermal recovery using PINN in the next section.

Table 1. The result of the static formation temperature estimation.

Field	SFT (°C)		R^2 value	
	Ascencio Method	**Horner Method**	**Ascencio Method**	**Horner Method**
MXCO1	283.30	253.42	0.98	0.96
MXCO2	282.71	251.30	0.97	0.95
SGIL	101.73	98.45	0.99	0.99
USAM	146.04	145.19	0.91	0.89
PHIL	178.81	171.43	0.95	0.97
JAPN	193.33	167.40	0.99	0.97
CH-A1	155.91	139.79	0.97	0.94
CH-A2	241.38	238.95	0.95	0.91
COST	64.01	59.85	0.89	0.92
ITAL	142.39	127.86	0.99	0.99

4.2 Thermal Recovery Model Result Using PINN

Several assumptions must be used to form a thermal recovery model using the PINN model. The boundary conditions are defined with an assumed farthest radius of 6.5 meters for each well, based on the requirement that the ratio between the farthest radius and the wellbore radius must be sufficiently large to satisfy the assumptions used in the SFT estimation. The 6.5-meter radius is considered appropriate for the current analysis. However, in future best practices, this value should ideally be determined through iterative trial and error to ensure greater accuracy and alignment with specific well conditions. The SFT value generated from the Horner and Ascencio methods will be used to model the thermal recovery model boundary conditions. Thermal diffusivity parameters in the initial scenario will be determined to model thermal recovery, as shown in Table 2.

Table 2. Thermal diffusivity parameters.

Zone	r_w (in.)	κ (W/m K)	ρ (kg/m3)	C_p (J/kg K)	D_t (m2/s)
observed zone	7	3.00	2500.00	900.00	1.33E-06

The total loss curve for each PINN training epoch can be seen in Figure 3. Based on the curve, the loss of PINN needs a few training iterations to converge. This curve will undoubtedly depend on the hyperparameter value of each part of the total loss discussed in Chapter 2.

To see the validation of existing data, Figure 4 shows the PINN prediction results on well heating up data when the radius is equal to the well radius. Based on Figure 4, the PINN prediction results still provide a representative value of the existing data. Figure 5 shows the cross-plot results between PINN predictions and original field data. Based on Figure 5, the deviation between the prediction and reference data is visually still in line so that it is representative.

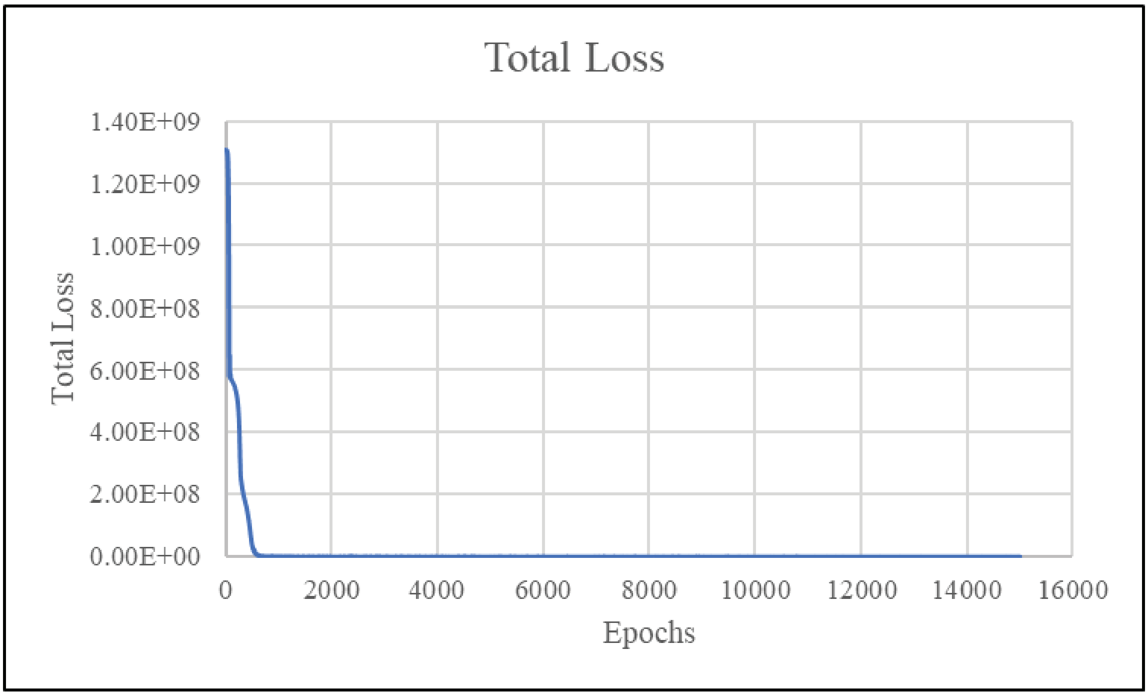

Figure 3. Total loss for PINN training.

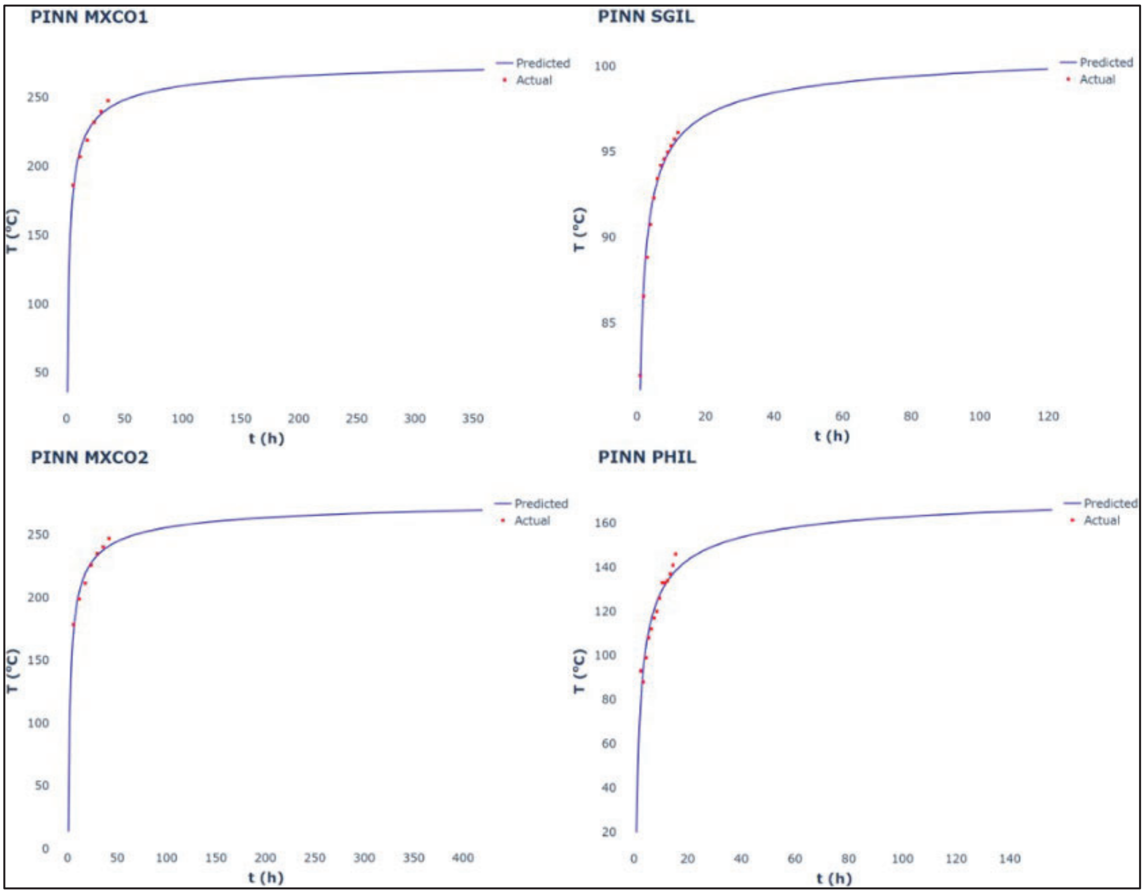

Figure 4. Timeseries data in comparison between PINN model predicted (blue lines) and actual field data (red markers).

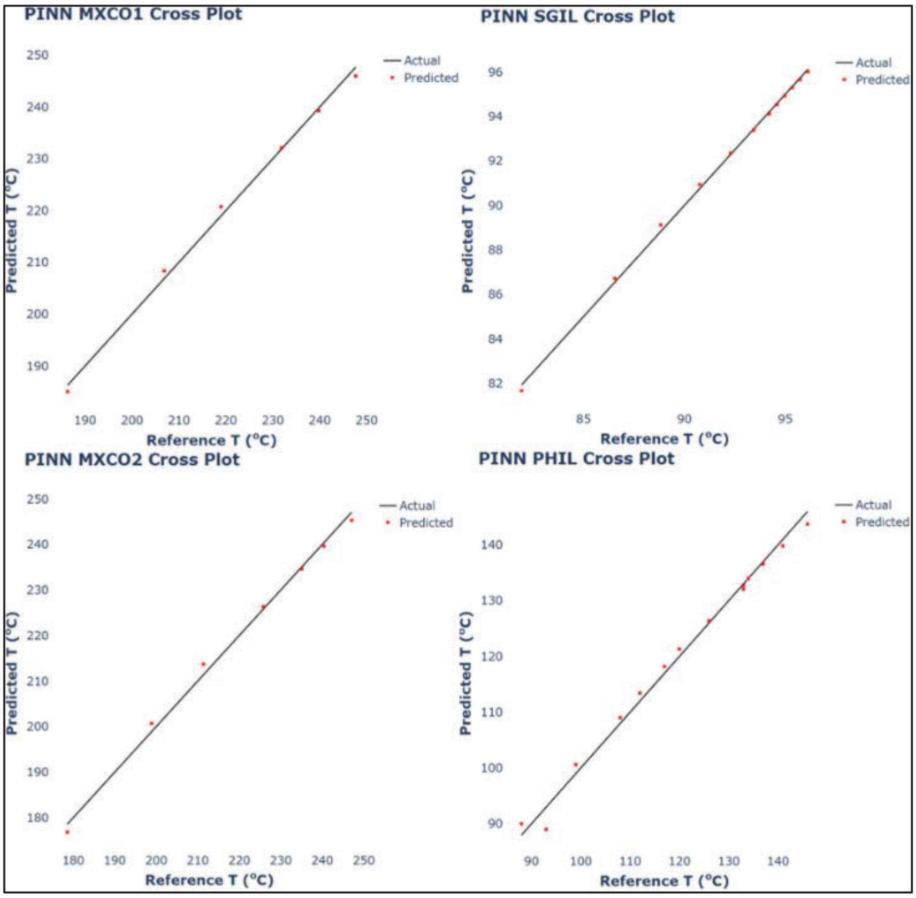

Figure 5. Cross-plot results to compare PINN model predicted and actual field data.

The results of each well modeling can be analyzed based on the plot between temperature and radius at each timestep, as well as the plot between temperature and time at each radius. The plot between temperature and radius at each timestep can be seen in Figure 6. Some wells such as SGIL, PHIL, and JAPN can describe the thermal recovery phase from time to time which is clean seen from the constant trend of each timestep in the few meters before the investigation radius which indicates that the assumption of the investigation radius used is appropriate for the well. Whereas in several other wells such as USAM, CH-A2, and KJ-31, the assumption of the investigation radius is too far and/or the delivery process is fast so that the thermal recovery process cannot be described as expected.

The plot between temperature and time at each radius can be seen in Figure 7. Based on these plots, thermal recovery modeling in almost every well can be seen quite well and as expected. Anomalies that can be found include, very fast heat propagation at several radii in the USAM and CH-A2 wells, and there is heat transmission from the well back to the surrounding radius which is indicated by a decrease in temperature at a certain radius from the beginning of the simulation time then heating up again where this indicates the assumption of inappropriate heat propagation parameters.

The results of thermal recovery modeling can also be depicted using two-dimensional illustrations with radial to cartesian axis conversion. Thermal recovery modeling that is well recorded in two-dimensional illustrations such as from the SGIL, MCCO1, and SHBE wells can be seen in Figure 8.

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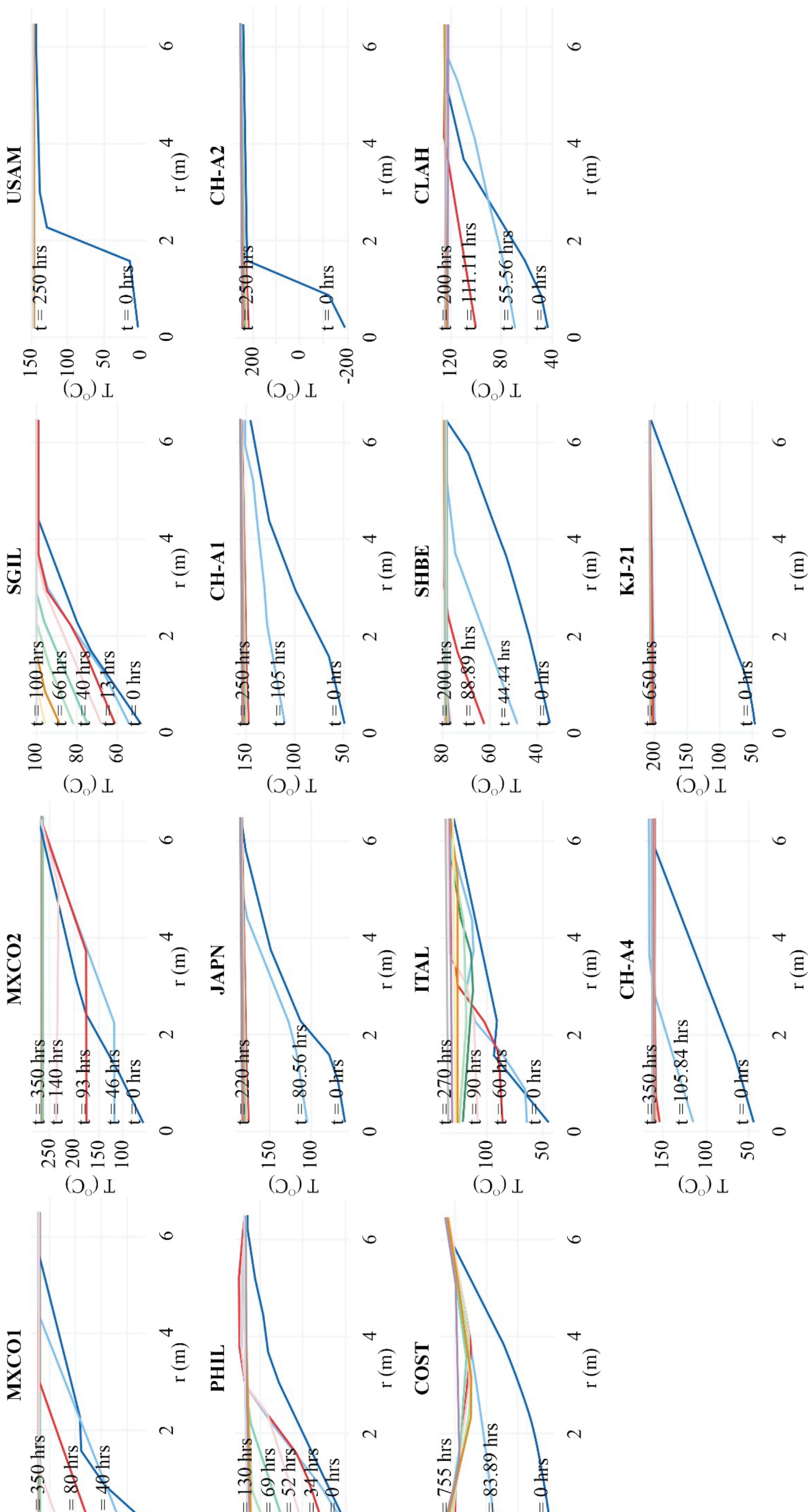

Figure 6. The plot between temperature and radius at each timestep.

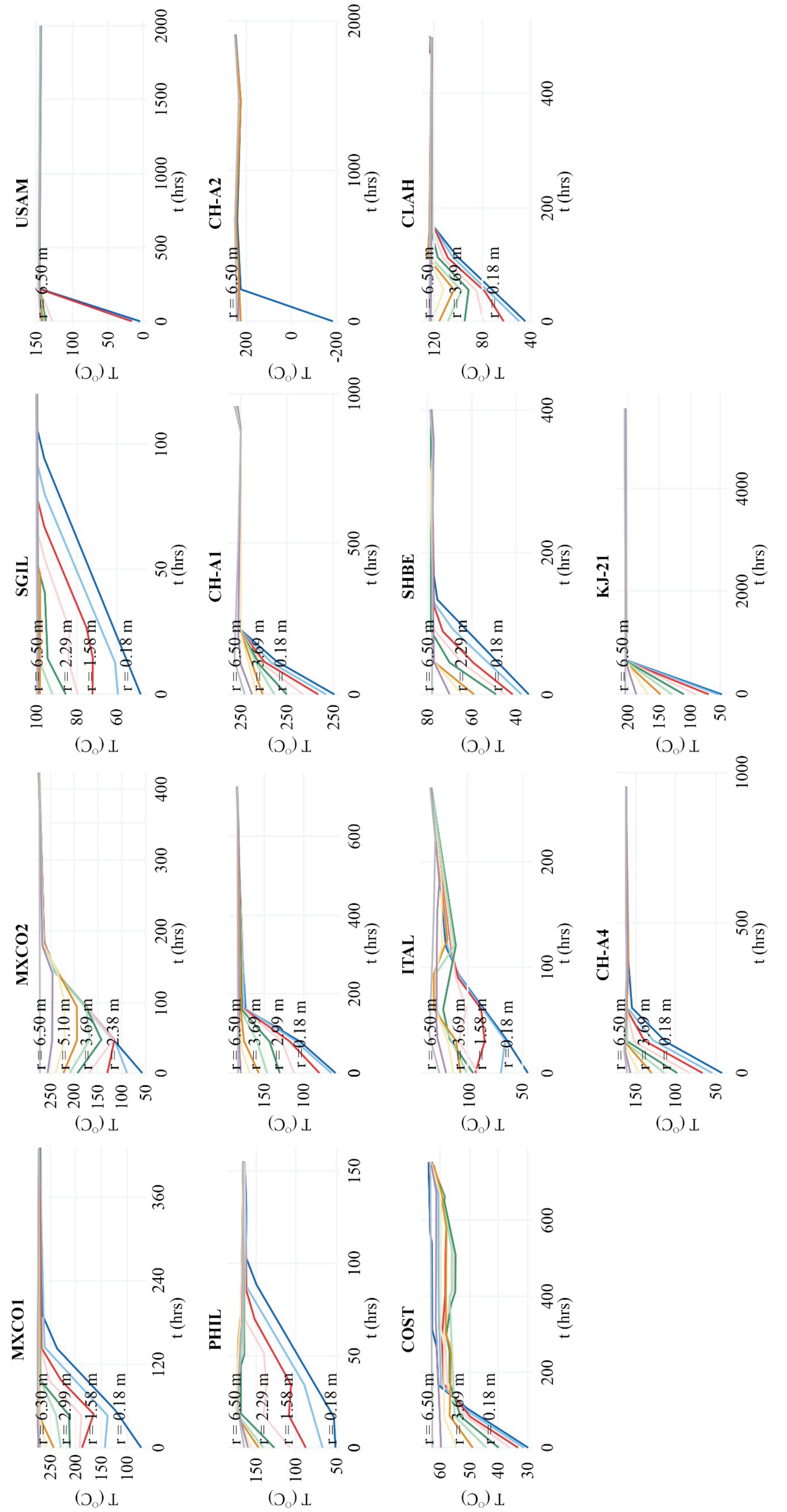

Figure 7. The plot between temperature and time at each radius.

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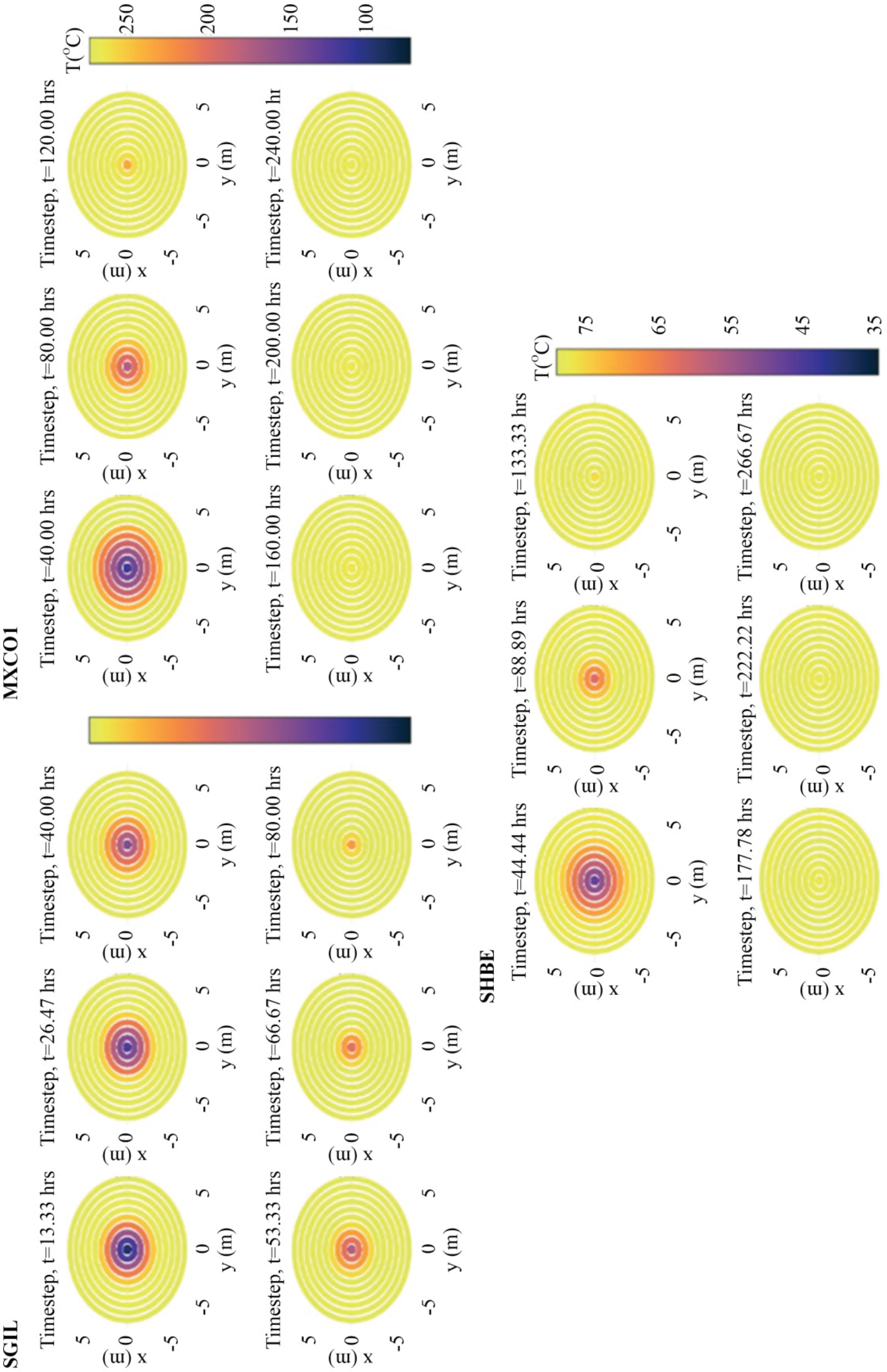

Figure 8. Two-dimensional illustrations for SGIL, MXCO1, and SHBE.

5. Conclusion and Recommendations

5.1 Conclusion

The successful application of Physics-Informed Neural Network (PINN) to model post-injection thermal recovery from geothermal wells has been achieved by employing the thermal radial diffusivity equation as the governing principle. It is imperative to predict the static formation temperature using the Horner and Ascencio analytical method to facilitate the modeling process using PINN, which serves as a necessary boundary condition. This modeling effort involved original data collected from various geothermal fields, extracted from a comprehensive review of existing literature. While the results of this modeling effort demonstrate the feasibility of employing PINNs to represent post-injection heat recovery in some of the well data utilized, the effectiveness of this modeling remains heavily reliant on the assumptions made regarding the investigation radius and the diffusivity parameters.

5.2 Recommendation

Modeling with a Physics-Informed Neural Network (PINN) can be enhanced into a Theory-guided Neural Network (TgNN) by incorporating additional elements like expert judgment or algorithm integration. This enhancement is necessary because the current model relies on trial and error to establish reasonable assumptions of investigation radius and diffusivity to improve this process, existing neural network algorithms such as backpropagation inverse modelling can be utilized. Furthermore, adopting probabilistic approaches such as Bayesian methods or Monte Carlo simulations may help reduce the reliance on trial and error. Additionally, strengthening the model's credibility can be achieved through a comparative analysis of its results with those derived from existing commercial software.

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